

Online Course Learning Progress Management System

Database Management System — Project Presentation

ER MODELING

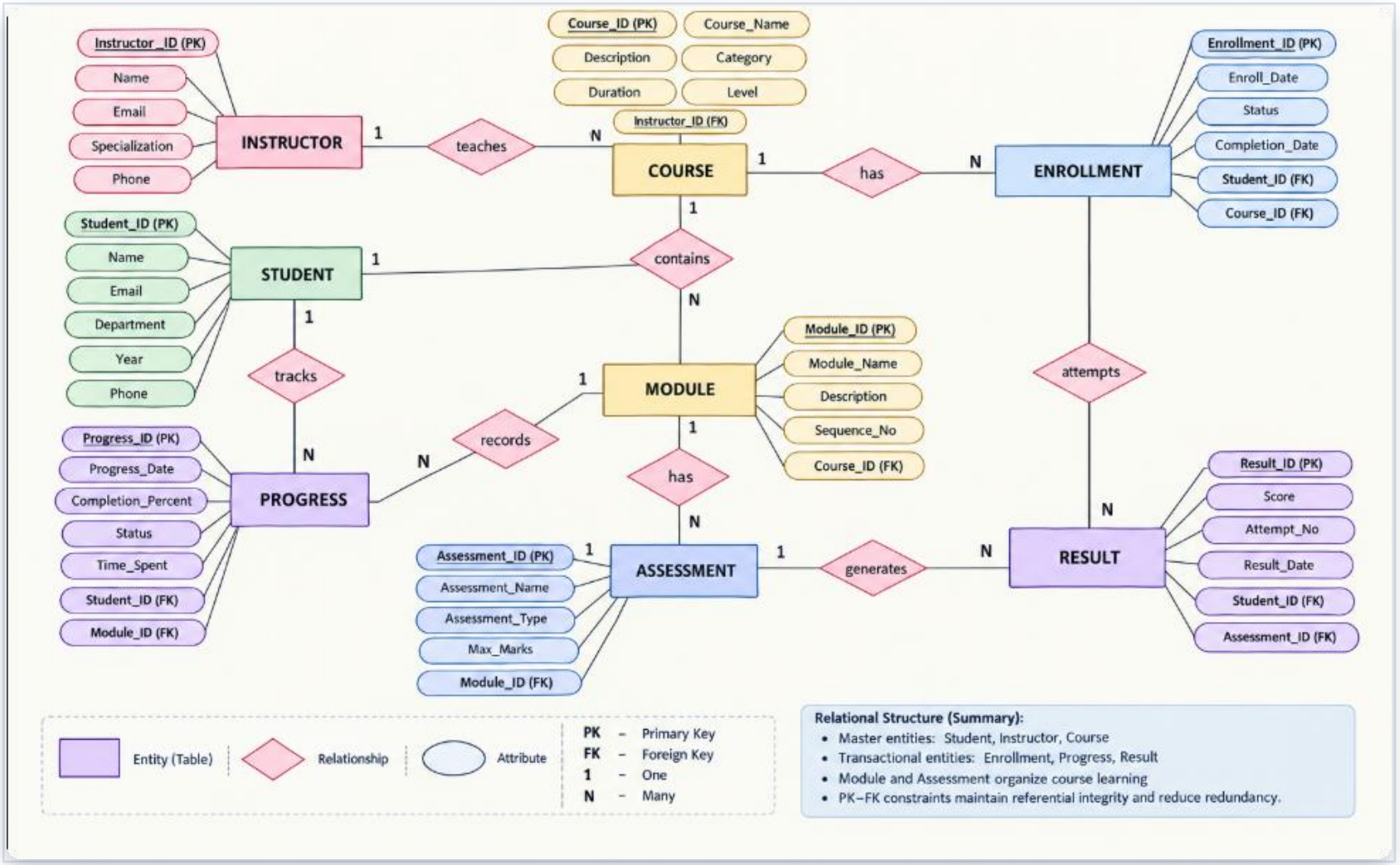
RELATIONAL DESIGN

SQL QUERIES

Built on MySQL · 8 Tables · Full Referential Integrity

Entity-Relationship Diagram

Eight entities model the full course-to-completion lifecycle, from enrollment to final results.



Database Design

Primary and foreign key relationships across all eight tables.

Table	Primary Key	Foreign Key
Student	Student_ID	—
Instructor	Instructor_ID	—
Course	Course_ID	Instructor_ID
Module	Module_ID	Course_ID
Assessment	Assessment_ID	Module_ID
Enrollment	Enrollment_ID	Student_ID, Course_ID
Progress	Progress_ID	Student_ID, Module_ID
Result	Result_ID	Student_ID, Assessment_ID

Master entity Organizing entity Transactional entity

Students Database

Core profile records for every learner on the platform.

The STUDENT table anchors the schema — every enrollment, progress record, and result traces back to a Student_ID.

```
SELECT * FROM STUDENT;
```

30
Total Students

3
Departments

1–3
Year Levels

Departments: Computer Science · Information Technology · Electronics

```
mysql> SELECT * FROM STUDENT;
```

Student_ID	Name	Email	Department	Year	Phone
1	Aarav Patel	aarav@univ.edu	Computer Science	1	555-1001
2	Aditi Sharma	aditi@univ.edu	Information Tech	2	555-1002
3	Rohan Gupta	rohan@univ.edu	Computer Science	1	555-1003
4	Sneha Reddy	sneha@univ.edu	Electronics	3	555-1004
5	Karan Singh	karan@univ.edu	Computer Science	2	555-1005
6	Priya Verma	priya@univ.edu	Information Tech	1	555-1006
7	Rahul Desai	rahul@univ.edu	Computer Science	1	555-1007
8	Ananya Nair	ananya@univ.edu	Electronics	2	555-1008
9	Vikram Joshi	vikram@univ.edu	Computer Science	3	555-1009
10	Neha Mehra	neha@univ.edu	Information Tech	1	555-1010
11	Arjun Das	arjun@univ.edu	Computer Science	2	555-1011
12	Kavya Iyer	kavya@univ.edu	Electronics	1	555-1012
13	Dev Kumar	dev@univ.edu	Information Tech	3	555-1013
14	Isha Rao	isha@univ.edu	Computer Science	1	555-1014
15	Samir Jain	samir@univ.edu	Electronics	2	555-1015
16	Maya Sen	maya@univ.edu	Computer Science	1	555-1016
17	Kabir Bose	kabir@univ.edu	Information Tech	3	555-1017
18	Tara Menon	tara@univ.edu	Computer Science	2	555-1018
19	Yash Khanna	yash@univ.edu	Electronics	1	555-1019
20	Sanya Kapoor	sanya@univ.edu	Information Tech	1	555-1020
21	Aryan Bhat	aryan@univ.edu	Computer Science	2	555-1021
22	Nisha Pillai	nisha@univ.edu	Electronics	3	555-1022
23	Dhruv Ahuja	dhruv@univ.edu	Computer Science	1	555-1023
24	Riya Ghosh	riya@univ.edu	Information Tech	2	555-1024
25	Omkar Nath	omkar@univ.edu	Computer Science	1	555-1025
26	Zara Sheikh	zara@univ.edu	Electronics	1	555-1026
27	Vedant Soni	vedant@univ.edu	Information Tech	2	555-1027
28	Mira Rajput	mira@univ.edu	Computer Science	3	555-1028
29	Armaan Malik	armaan@univ.edu	Electronics	1	555-1029
30	Pooja Hegde	pooja@univ.edu	Information Tech	2	555-1030

SQL Query — Filtering Records

Retrieve every first-year student, regardless of department.

```
mysql> SELECT Name, Department, Year FROM STUDENT WHERE Year = 1;
+--------------+------------------+------+
| Name         | Department       | Year |
+--------------+------------------+------+
| Aarav Patel  | Computer Science |    1 |
| Rohan Gupta  | Computer Science |    1 |
| Priya Verma  | Information Tech |    1 |
| Rahul Desai  | Computer Science |    1 |
| Neha Mehra   | Information Tech |    1 |
| Kavya Iyer   | Electronics      |    1 |
| Isha Rao     | Computer Science |    1 |
| Maya Sen     | Computer Science |    1 |
| Yash Khanna  | Electronics      |    1 |
| Sanya Kapoor | Information Tech |    1 |
| Dhruv Ahuja  | Computer Science |    1 |
| Omkar Nath   | Computer Science |    1 |
| Zara Sheikh  | Electronics      |    1 |
| Armaan Malik | Electronics      |    1 |
+--------------+------------------+------+
```

The WHERE clause narrows the STUDENT table to a single condition — Year = 1 — surfacing new learners across every department.

```
SELECT Name, Department, Year
FROM STUDENT
WHERE Year = 1;
```

14 first-year students matched the filter

Useful for advisor outreach, orientation lists, and first-year cohort tracking.

SQL Query — Joining Tables

Combine Instructor and Course records through a shared key.

An INNER JOIN links INSTRUCTOR and COURSE on Instructor_ID, revealing exactly which courses each instructor teaches.

```
SELECT i.Name AS Instructor_Name,
       c.Course_Name, c.Category
FROM INSTRUCTOR i
JOIN COURSE c
  ON i.Instructor_ID = c.Instructor_ID;
```

```
mysql> SELECT
    ->     i.Name AS Instructor_Name,
    ->     c.Course_Name,
    ->     c.Category
    -> FROM INSTRUCTOR i
    -> JOIN COURSE c ON i.Instructor_ID = c.Instructor_ID;
+-----------------+------------------+------------------+
| Instructor_Name | Course_Name      | Category         |
+-----------------+------------------+------------------+
| Dr. Alan Turing | Intro to Python  | Programming      |
| Dr. Alan Turing | Database Systems | Computer Science |
+-----------------+------------------+------------------+
```

2 courses taught by Dr. Alan Turing

This join pattern powers every “who teaches what” report in the platform.